

Chemistry worked solutions and marking guide

Sylligence mock external examination

Papers 1 and 2 - combined guide (110 marks)

How to use this guide

Each response shows a complete chemically valid pathway. The worked stages explain the relevant principle, evidence, calculation or inference. The detailed marking scheme separately identifies the evidence required for every mark, including equivalent valid responses where appropriate.

Assessment objectives

This pair assesses description and explanation of chemical concepts, application of understanding, analysis and interpretation of evidence, and evaluation of chemical claims and conclusions across Units 3 and 4.



Paper 1 multiple choice key and reasoning

Question 1

B

WORKED REASONING

At dynamic equilibrium, both reactions continue, but their rates are equal. Concentrations are constant, not necessarily equal.

Why the other options do not follow

- A Constant concentrations do not require equal concentrations, and both reactions remain active.
- C A non-zero reverse rate is essential to dynamic equilibrium.
- D Particles continue moving and colliding in the closed system.

Question 2

D

WORKED REASONING

The group -OH is the hydroxyl functional group; a molecule containing it in this context belongs to the alcohol class.

Why the other options do not follow

- A The functional-group and class names are reversed.
- B A carbonyl group contains a carbon–oxygen double bond, which is not highlighted.
- C A carboxyl group contains both C=O and O-H on the same carbon.

Question 3

A

WORKED REASONING

Pure solids are omitted from a heterogeneous equilibrium expression, so $K_c = [\text{CO}_2]$.

Why the other options do not follow

- B Concentrations of pure solids do not appear in the expression.
- C The pure solid CaO is omitted.
- D Carbon dioxide is a product, so its concentration is in the numerator.

Question 4

C

WORKED REASONING

Each carbon of the double bond has one CH_3 group and one H atom. The two CH_3 groups are on opposite sides, so the molecule is trans-but-2-ene.

Why the other options do not follow

- A But-1-ene does not have two different substituents on both double-bond carbons and therefore has no cis/trans pair.
- B The double bond is between carbons 2 and 3, not carbon 1.
- D The two methyl groups are on opposite, not the same, sides of the double bond.

Paper 1 multiple choice key and reasoning

Question 5

B

WORKED REASONING

Because the analytical concentrations are equal, the lower pH and higher conductivity show that P ionises to a greater extent and is the stronger acid.

Why the other options do not follow

- A At equal concentration, the higher pH indicates less ionisation and therefore the weaker acid.
- C The question states that the analytical concentrations are equal.
- D Lower conductivity is consistent with fewer mobile ions and less extensive ionisation.

Question 6

A

WORKED REASONING

Restoring the carbon–carbon double bond in the repeat-unit backbone gives $\text{CF}_2=\text{CF}_2$, tetrafluoroethene.

Why the other options do not follow

- B An alkane has no carbon–carbon double bond to open during addition polymerisation.
- C Ethene would form a repeat unit containing only carbon and hydrogen.
- D A saturated fluoroalkane has no carbon–carbon double bond to open during addition polymerisation.

Question 7

D

WORKED REASONING

$$Q = \frac{(0.500)^2}{(0.200)(0.200)} = 6.25. \text{ Since } Q < K_c, \text{ the system proceeds forward to form more HI.}$$

Why the other options do not follow

- A Q is 6.25, not 50.0.
- B The calculated reaction quotient is less than, not greater than, K_c .
- C When $Q < K_c$, products must form, so the net direction is forward.

Question 8

C

WORKED REASONING

H_2 is limiting. It forms 3.00 mol of NH_3 , or 51.12 g. The yield is $38.3/51.12 \times 100 = 74.9\%$.

Why the other options do not follow

- A This incorrectly uses a 1 : 1 hydrogen-to-ammonia mole ratio.
- B This incorrectly treats nitrogen, rather than hydrogen, as the limiting reagent.
- D This reverses the actual-to-theoretical mass ratio.

Paper 1 multiple choice key and reasoning

Question 9

A

WORKED REASONING

At $\text{pH} = \text{p}K_a$, the acid and conjugate-base forms of the indicator have equal concentrations.

Why the other options do not follow

- B The indicator equilibrium does not require hydrogen-ion concentration to equal indicator concentration.
- C Both coloured forms are present at the midpoint.
- D The midpoint depends on the indicator $\text{p}K_a$, not on neutral pH.

Question 10

B

WORKED REASONING

Markovnikov addition places H on the terminal carbon and Br on carbon 2, giving 2-bromopentane.

Why the other options do not follow

- A This places Br on the less substituted terminal carbon.
- C Markovnikov addition places Br on carbon 2, not carbon 3.
- D Addition of HBr does not introduce a methyl branch into the five-carbon chain.

Question 11

C

WORKED REASONING

Silver ions are reduced and iron is oxidised. $E_{\text{cell}}^{\circ} = 0.80 - (-0.44) = +1.24 \text{ V}$.

Why the other options do not follow

- A A spontaneous galvanic cell has a positive standard potential in the written direction.
- B This subtracts the magnitudes rather than subtracting the oxidation-half-cell reduction potential.
- D This incorrectly doubles the silver reduction potential when balancing electrons and omits the iron half-cell contribution.

Question 12

D

WORKED REASONING

The very broad acid O–H absorption together with a carbonyl absorption identifies a carboxylic acid.

Why the other options do not follow

- A An alkene does not produce this broad O–H absorption and carbonyl combination.
- B An alcohol O–H band is not normally accompanied by the required carbonyl band.
- C An amine N–H absorption is narrower and does not account for the carbonyl band.

Paper 1 multiple choice key and reasoning

Question 13

B

WORKED REASONING

At the positive electrode, water/hydroxide is preferentially oxidised in dilute sodium chloride, producing oxygen.

Why the other options do not follow

- A Hydrogen forms by reduction at the negative electrode.
- C Sodium metal is not deposited from an aqueous solution because water is reduced preferentially.
- D Chlorine is favoured at the anode for concentrated chloride, not the stated dilute solution.

Question 14

A

WORKED REASONING

Increasing chain length increases electron number, polarisability and contact surface, strengthening dispersion forces and raising boiling point.

Why the other options do not follow

- B Alkanes do not contain groups that hydrogen-bond with one another.
- C Boiling overcomes intermolecular forces, not covalent bonds within molecules.
- D Alkanes are nonpolar, and the data show the longer chains have the higher boiling points.

Question 15

C

WORKED REASONING

If the molar solubility is s , then $K_{sp} = (2s)^2(s) = 4s^3$. Hence $s = (8.0 \times 10^{-12}/4)^{1/3} = 1.3 \times 10^{-4}$ M.

Why the other options do not follow

- A This divides K_{sp} by 4 but omits the required cube root.
- B This incorrectly takes $\sqrt{K_{sp}}$ rather than solving $K_{sp} = 4s^3$.
- D This omits the factor of 4 from $[\text{Ag}^+]^2 = (2s)^2$.

Question 16

D

WORKED REASONING

Because the buffer pH is above the isoelectric point, the amino acid has a net negative charge and migrates towards the positive electrode.

Why the other options do not follow

- A A net positive charge occurs when the buffer pH is below the isoelectric point.
- B The net charge is zero only near the amino acid's own isoelectric point.
- C A negatively charged ion is attracted to the positive electrode.

Paper 1 multiple choice key and reasoning

Question 17

A

WORKED REASONING

The weak acid component consumes much of the added hydroxide, forming acetate and water. The pH therefore rises only slightly, not by the amount it would in water.

Why the other options do not follow

- B A buffer resists pH change; it does not keep pH exactly constant.
- C The conjugate-pair concentrations change slightly as the added base is consumed.
- D Sodium ions are spectator ions and do not form sodium metal in water.

Question 18

C

WORKED REASONING

The side chains from left to right are CH_3 (Ala), H (Gly) and CH_2OH (Ser), giving Ala–Gly–Ser.

Why the other options do not follow

- A This reverses the sequence direction and swaps the terminal residues.
- B The first residue has a methyl side chain, not H.
- D The middle residue has H as its side chain, so it is glycine.

Question 19

D

WORKED REASONING

For the reverse reaction, $E_a = 125 - 75 = 50 \text{ kJ mol}^{-1}$.

Why the other options do not follow

- A This is the forward enthalpy change, $75 - 40$.
- B This is the reactant energy level, not an energy barrier.
- C This is the forward activation energy, $125 - 40$.

Question 20

B

WORKED REASONING

Reduction occurs at the cathode. Oxygen gains electrons and combines with hydrogen ions to form water.

Why the other options do not follow

- A This is the reverse, oxidation direction.
- C This is the alkaline anode half-equation.
- D Hydrogen ions are not reduced to hydrogen in an operating hydrogen–oxygen fuel cell cathode.

PAPER 1 - QUESTION 21

(3 marks)

Transition-metal oxidation states are represented by a signed number, while ionic compound names use a Roman numeral.

- a) Determine the oxidation state of cobalt in Co_2O_3 . [1 mark]

WORKED SOLUTION

STEP 1 Oxygen has oxidation state -2 . Therefore $2x + 3(-2) = 0$, giving $x = +3$.

DETAILED MARKING SCHEME

MARK	EVIDENCE REQUIRED	VALUE
Mark 1	Determines the cobalt oxidation state as $+3$.	1 mark

Award credit for an equivalent chemically valid method when the same evidence is demonstrated. A correct answer without required supporting working may not earn every mark.

- b) Name Co_2O_3 . [1 mark]

WORKED SOLUTION

STEP 1 The compound is cobalt(III) oxide.

DETAILED MARKING SCHEME

MARK	EVIDENCE REQUIRED	VALUE
Mark 1	Names the compound cobalt(III) oxide, with the Roman numeral in parentheses.	1 mark

Award credit for an equivalent chemically valid method when the same evidence is demonstrated. A correct answer without required supporting working may not earn every mark.

- c) For $2\text{Co}^{3+}(\text{aq}) + \text{Sn}^{2+}(\text{aq}) \rightarrow 2\text{Co}^{2+}(\text{aq}) + \text{Sn}^{4+}(\text{aq})$, identify the oxidising agent. [1 mark]

WORKED SOLUTION

STEP 1 Co^{3+} gains electrons to form Co^{2+} , so Co^{3+} is the oxidising agent.

DETAILED MARKING SCHEME

MARK	EVIDENCE REQUIRED	VALUE
Mark 1	Identifies $\text{Co}^{3+}(\text{aq})$ as the oxidising agent. Also accept: cobalt(III) ions	1 mark

Award credit for an equivalent chemically valid method when the same evidence is demonstrated. A correct answer without required supporting working may not earn every mark.

PAPER 1 - QUESTION 22

(4 marks)

The table gives physical-property data for four organic compounds.

Compound	Class	Molar mass (g mol ⁻¹)	Boiling point (°C)	Water solubility
but-1-ene	alkene	56.1	−6	very low
pentane	alkane	72.2	36	very low
butan-1-ol	alcohol	74.1	118	moderate
propanoic acid	carboxylic acid	74.1	141	very high

Physical properties

- a) Determine the most volatile compound. Justify your answer using the data.

[2 marks]

WORKED SOLUTION

STEP 1 But-1-ene has the lowest boiling point, −6 °C.

STEP 2 A lower boiling point means a greater tendency to enter the gas phase, so but-1-ene is the most volatile.

DETAILED MARKING SCHEME

MARK	EVIDENCE REQUIRED	VALUE
Mark 1	Identifies but-1-ene.	1 mark
Mark 2	Uses its lowest boiling point to justify greatest volatility.	1 mark

Award credit for an equivalent chemically valid method when the same evidence is demonstrated. A correct answer without required supporting working may not earn every mark.

- b) Explain why butan-1-ol has a higher boiling point than pentane even though their molar masses are similar.

[2 marks]

WORKED SOLUTION

STEP 1 Both substances experience dispersion forces.

STEP 2 Butan-1-ol molecules also form hydrogen bonds through their hydroxyl groups, whereas pentane molecules do not. More energy is therefore required to separate butan-1-ol molecules.

DETAILED MARKING SCHEME

MARK	EVIDENCE REQUIRED	VALUE
Mark 1	Identifies hydrogen bonding between butan-1-ol molecules.	1 mark
Mark 2	Links the stronger intermolecular attraction to the higher boiling point.	1 mark

Award credit for an equivalent chemically valid method when the same evidence is demonstrated. A correct answer without required supporting working may not earn every mark.

PAPER 1 - QUESTION 23

(4 marks)

A saturated calcium fluoride solution is described by $\text{CaF}_2(\text{s}) \rightleftharpoons \text{Ca}^{2+}(\text{aq}) + 2\text{F}^{-}(\text{aq})$. At 25°C , the saturated solution has $[\text{F}^{-}] = 4.80 \times 10^{-4} \text{ M}$.

- a) Calculate K_{sp} for calcium fluoride.

[2 marks]

WORKED SOLUTION

STEP 1 Stoichiometry gives $[\text{Ca}^{2+}] = \frac{1}{2}[\text{F}^{-}] = 2.40 \times 10^{-4} \text{ M}$.

STEP 2 $K_{sp} = [\text{Ca}^{2+}][\text{F}^{-}]^2 = (2.40 \times 10^{-4})(4.80 \times 10^{-4})^2 = 5.53 \times 10^{-11} \text{ mol}^3 \text{ L}^{-3}$.

DETAILED MARKING SCHEME

MARK	EVIDENCE REQUIRED	VALUE
Mark 1	Determines $[\text{Ca}^{2+}] = 2.40 \times 10^{-4} \text{ M}$.	1 mark
Mark 2	Obtains $K_{sp} = 5.53 \times 10^{-11} \text{ mol}^3 \text{ L}^{-3}$. Also accept: $5.5 \times 10^{-11} \text{ mol}^3 \text{ L}^{-3}$ to two significant figures	1 mark

Award credit for an equivalent chemically valid method when the same evidence is demonstrated. A correct answer without required supporting working may not earn every mark.

- b) A small amount of calcium chloride is added without a significant change in volume. Predict whether a precipitate forms. Explain your reasoning.

[2 marks]

WORKED SOLUTION

STEP 1 Adding calcium chloride increases $[\text{Ca}^{2+}]$, so the ion product $Q = [\text{Ca}^{2+}][\text{F}^{-}]^2$ becomes greater than K_{sp} .

STEP 2 The equilibrium shifts left and $\text{CaF}_2(\text{s})$ precipitates until $Q = K_{sp}$ again.

DETAILED MARKING SCHEME

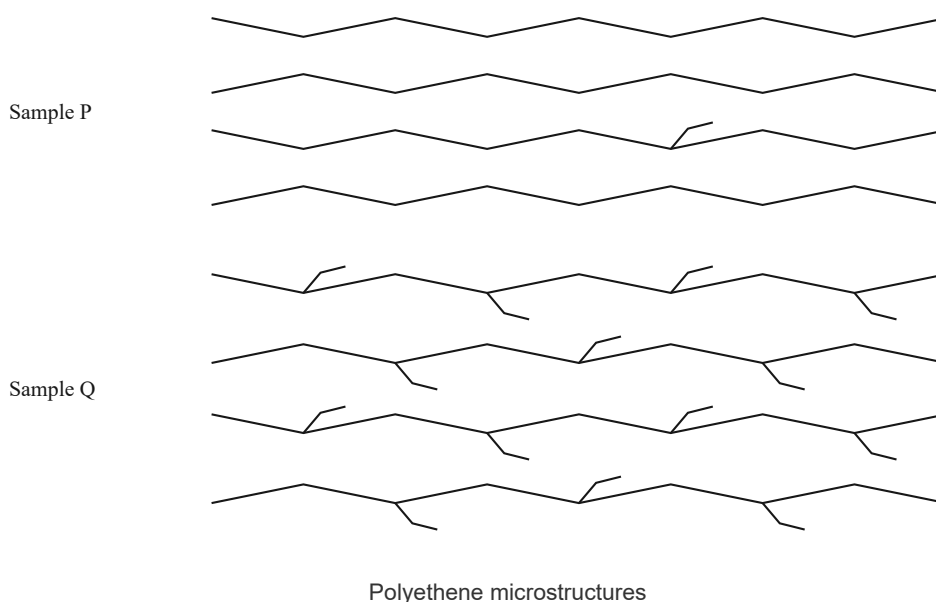
MARK	EVIDENCE REQUIRED	VALUE
Mark 1	Determines that $Q > K_{sp}$ after $[\text{Ca}^{2+}]$ increases.	1 mark
Mark 2	Predicts calcium fluoride precipitation/a left shift.	1 mark

Award credit for an equivalent chemically valid method when the same evidence is demonstrated. A correct answer without required supporting working may not earn every mark.

PAPER 1 - QUESTION 24

(4 marks)

Two polyethene samples have the chain arrangements shown.



- a) Identify the sample that represents HDPE and the sample that represents LDPE.

[2 marks]

WORKED SOLUTION

STEP 1 Sample P has little branching and represents HDPE. Sample Q has extensive branching and represents LDPE.

DETAILED MARKING SCHEME

MARK	EVIDENCE REQUIRED	VALUE
Mark 1	Identifies P as HDPE.	1 mark
Mark 2	Identifies Q as LDPE.	1 mark

Award credit for an equivalent chemically valid method when the same evidence is demonstrated. A correct answer without required supporting working may not earn every mark.

- b) Explain why sample P has a greater density and generally greater strength than sample Q.

[2 marks]

WORKED SOLUTION

STEP 1 The less-branched chains of P pack more closely than the highly branched chains of Q.

STEP 2 Closer packing produces more effective intermolecular contact and stronger cumulative dispersion forces, increasing density and strength.

DETAILED MARKING SCHEME

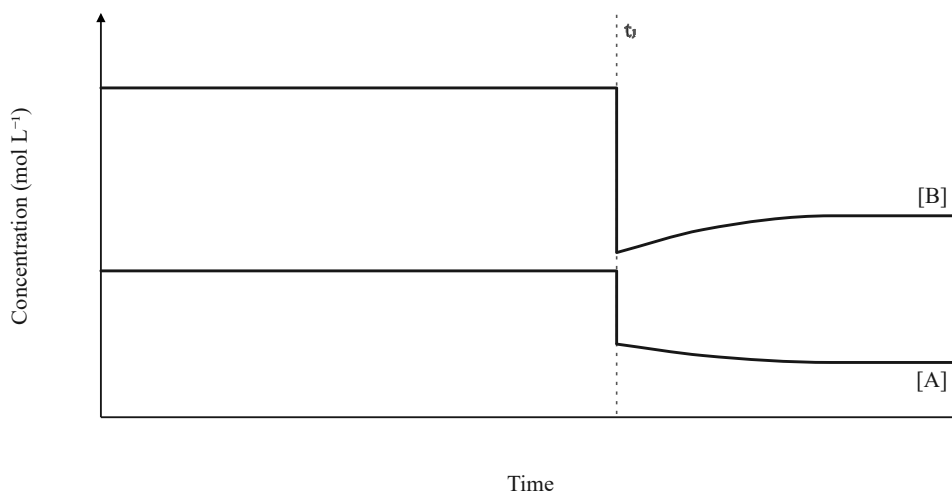
MARK	EVIDENCE REQUIRED	VALUE
Mark 1	Links low branching to closer chain packing.	1 mark
Mark 2	Links closer packing/stronger dispersion interactions to greater density and strength.	1 mark

Award credit for an equivalent chemically valid method when the same evidence is demonstrated. A correct answer without required supporting working may not earn every mark.

PAPER 1 - QUESTION 25

(4 marks)

The graph shows the concentrations in the equilibrium $A(g) \rightleftharpoons 2B(g)$. A single change is applied at t_1 while temperature remains constant.



Concentration response to a change at t_1

- a) Identify the change applied at t_1 .

[1 mark]

WORKED SOLUTION

STEP 1 Both concentrations decrease instantaneously by the same factor, so the vessel volume was increased.

DETAILED MARKING SCHEME

MARK	EVIDENCE REQUIRED	VALUE
Mark 1	Identifies an increase in volume/decrease in pressure.	1 mark

Award credit for an equivalent chemically valid method when the same evidence is demonstrated. A correct answer without required supporting working may not earn every mark.

- b) Explain the instantaneous changes in both concentrations at t_1 .

[1 mark]

WORKED SOLUTION

STEP 1 Increasing the volume dilutes both gases immediately, so both concentrations fall before any equilibrium shift occurs.

DETAILED MARKING SCHEME

MARK	EVIDENCE REQUIRED	VALUE
Mark 1	Explains that dilution from the volume increase causes the instantaneous fall in both concentrations.	1 mark

Award credit for an equivalent chemically valid method when the same evidence is demonstrated. A correct answer without required supporting working may not earn every mark.

PAPER 1 - QUESTION 25 - CONTINUED

(4 marks)

- c) Explain the subsequent changes as the system establishes a new equilibrium.

[2 marks]

WORKED SOLUTION

STEP 1 The lower pressure favours the side with more gas particles, which is the product side.

STEP 2 A is consumed and B forms until the forward and reverse rates become equal at new plateaux; K_c is unchanged because temperature is unchanged.

DETAILED MARKING SCHEME

MARK	EVIDENCE REQUIRED	VALUE
Mark 1	Uses gas-particle numbers to predict a shift to the right.	1 mark
Mark 2	Links the shift to decreasing A/increasing B relative to their immediate post-change values, with K_c unchanged.	1 mark

Award credit for an equivalent chemically valid method when the same evidence is demonstrated. A correct answer without required supporting working may not earn every mark.

PAPER 1 - QUESTION 26

(4 marks)

But-2-ene has the molecular formula C_4H_8 .

- a) Draw both geometrical isomers of but-2-ene.

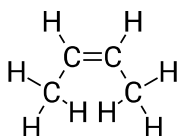
[2 marks]

WORKED SOLUTION

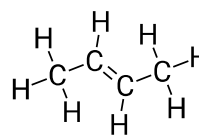
STEP 1 Draw cis-but-2-ene with the two methyl groups on the same side of the double bond.

STEP 2 Draw trans-but-2-ene with the methyl groups on opposite sides.

cis-but-2-ene



trans-but-2-ene



Model structures

DETAILED MARKING SCHEME

MARK	EVIDENCE REQUIRED	VALUE
Mark 1	Draws cis-but-2-ene with correct connectivity and geometry.	1 mark
Mark 2	Draws trans-but-2-ene with correct connectivity and geometry.	1 mark

Award credit for an equivalent chemically valid method when the same evidence is demonstrated. A correct answer without required supporting working may not earn every mark.

- b) Describe the result when either isomer is shaken with bromine water. Explain the observation.

[2 marks]

WORKED SOLUTION

STEP 1 The orange/brown bromine water is decolourised.

STEP 2 Bromine adds across the carbon-carbon double bond to form a dibromoalkane.

DETAILED MARKING SCHEME

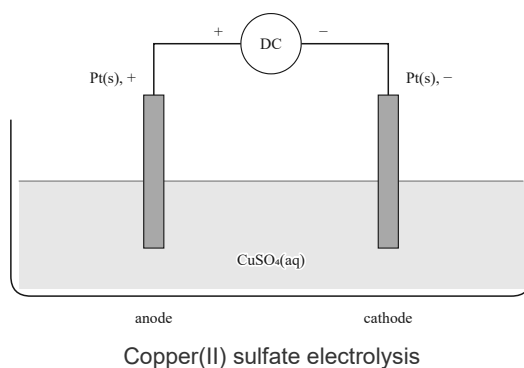
MARK	EVIDENCE REQUIRED	VALUE
Mark 1	States that bromine water is decolourised.	1 mark
Mark 2	Explains that bromine undergoes addition across the $C=C$ bond.	1 mark

Award credit for an equivalent chemically valid method when the same evidence is demonstrated. A correct answer without required supporting working may not earn every mark.

PAPER 1 - QUESTION 27

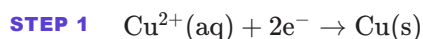
(4 marks)

Aqueous copper(II) sulfate is electrolysed using inert platinum electrodes. A current of 1.50 A passes for 1930 s.



- a) Write the cathode half-equation. Include states. [1 mark]

WORKED SOLUTION



DETAILED MARKING SCHEME

MARK	EVIDENCE REQUIRED	VALUE
Mark 1	Writes $\text{Cu}^{2+}(\text{aq}) + 2\text{e}^{-} \rightarrow \text{Cu}(\text{s})$ with correct charge, electrons and states.	1 mark

Award credit for an equivalent chemically valid method when the same evidence is demonstrated. A correct answer without required supporting working may not earn every mark.

- b) Calculate the mass of copper deposited. [3 marks]

WORKED SOLUTION

STEP 1 $q = It = (1.50)(1930) = 2895 \text{ C}$.

STEP 2 $n(\text{e}^{-}) = q/F = 2895/96500 = 0.0300 \text{ mol}$, so $n(\text{Cu}) = 0.0150 \text{ mol}$.

STEP 3 $m = nM = (0.0150)(63.55) = 0.953 \text{ g}$.

DETAILED MARKING SCHEME

MARK	EVIDENCE REQUIRED	VALUE
Mark 1	Calculates charge $q = 2895 \text{ C}$.	1 mark
Mark 2	Uses the two-electron ratio to obtain $n(\text{Cu}) = 0.0150 \text{ mol}$.	1 mark
Mark 3	Obtains a deposited copper mass of 0.953 g. Also accept: 0.95 g to two significant figures	1 mark

Award credit for an equivalent chemically valid method when the same evidence is demonstrated. A correct answer without required supporting working may not earn every mark.

PAPER 1 - QUESTION 28

(4 marks)

Ethanol is produced by hydration of ethene: $\text{C}_2\text{H}_4(\text{g}) + \text{H}_2\text{O}(\text{g}) \rightarrow \text{C}_2\text{H}_5\text{OH}(\text{g})$. A plant must collect 18.4 kg of ethanol from a process operating at 80.0% yield.

Calculate the minimum mass of ethene that must be supplied to the process. Assume water is in excess. [4 marks]

WORKED SOLUTION

STEP 1 The theoretical ethanol mass required is $18.4/0.800 = 23.0 \text{ kg}$.

STEP 2 $n(\text{C}_2\text{H}_5\text{OH}) = 23000/46.08 = 499 \text{ mol}$.

STEP 3 The equation has a 1 : 1 ethene-to-ethanol mole ratio, so $n(\text{C}_2\text{H}_4) = 499 \text{ mol}$.

STEP 4 $m(\text{C}_2\text{H}_4) = (499)(28.06) = 1.40 \times 10^4 \text{ g} = 14.0 \text{ kg}$.

DETAILED MARKING SCHEME

MARK	EVIDENCE REQUIRED	VALUE
Mark 1	Converts actual yield to theoretical ethanol mass, 23.0 kg.	1 mark
Mark 2	Converts theoretical ethanol mass to approximately 499 mol.	1 mark
Mark 3	Applies the 1 : 1 mole ratio to ethene.	1 mark
Mark 4	Obtains approximately 14.0 kg of ethene.	1 mark

Award credit for an equivalent chemically valid method when the same evidence is demonstrated. A correct answer without required supporting working may not earn every mark.

PAPER 1 - QUESTION 29

(4 marks)

Four 0.0100 M solutions are hydrochloric acid, ethanoic acid, sodium hydroxide and ammonia. The labels have been removed. Use all relevant evidence to identify the solutions.

Solution	pH	Conductivity
P	2.00	high
Q	3.38	low
R	12.00	high
S	10.62	low

Measurements for four unknown solutions

Match P, Q, R and S to the four substances. Explain your assignments using pH, conductivity and degree of ionisation at equilibrium. [4 marks]

WORKED SOLUTION

- STEP 1** P is hydrochloric acid: its pH is 2.00 at 0.0100 M and it has high conductivity, consistent with a strong acid.
- STEP 2** Q is ethanoic acid: it is acidic but has a higher pH and lower conductivity than P, consistent with partial ionisation.
- STEP 3** R is sodium hydroxide: its pH is 12.00 and its conductivity is high, consistent with a strong base.
- STEP 4** S is ammonia: it is basic but has lower conductivity and a less extreme pH than R, consistent with a weak base.

DETAILED MARKING SCHEME

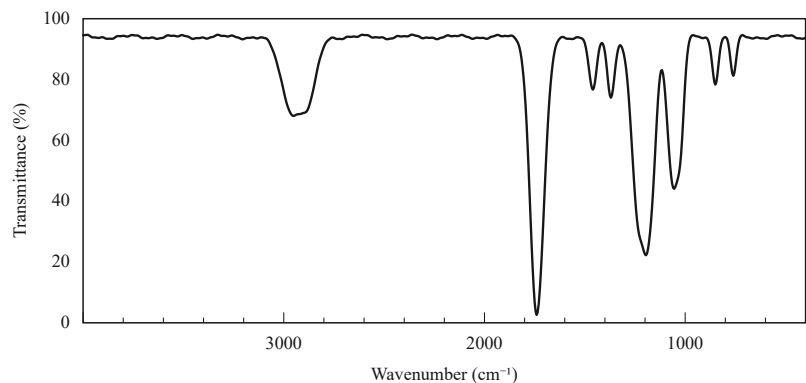
MARK	EVIDENCE REQUIRED	VALUE
Mark 1	Identifies P as hydrochloric acid using acidic pH and strong-electrolyte evidence.	1 mark
Mark 2	Identifies Q as ethanoic acid using partial-ionisation evidence.	1 mark
Mark 3	Identifies R as sodium hydroxide using basic pH and strong-electrolyte evidence.	1 mark
Mark 4	Identifies S as ammonia using weak-base evidence.	1 mark

Award credit for an equivalent chemically valid method when the same evidence is demonstrated. A correct answer without required supporting working may not earn every mark.

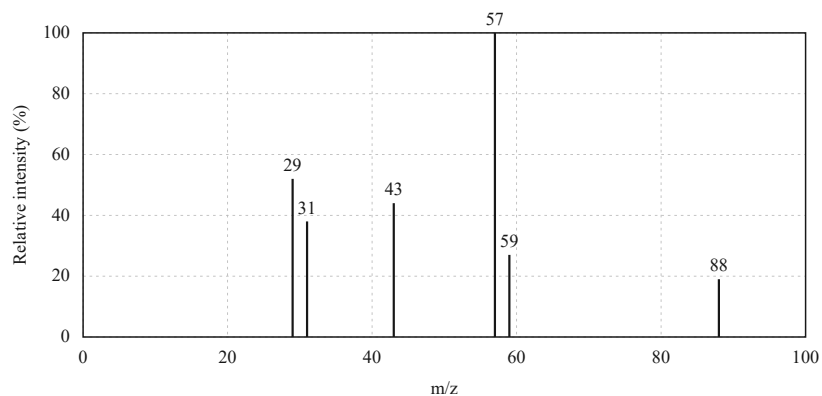
PAPER 2 - QUESTION 1

(6 marks)

An organic liquid has empirical formula C_2H_4O . It was produced by reacting methanol with a carboxylic acid. Its infrared and mass spectra are shown.



Infrared spectrum



Mass spectrum

- a) Identify the compound class and justify your answer using the infrared spectrum.

[2 marks]

WORKED SOLUTION

STEP 1 The strong absorption near 1740 cm^{-1} indicates a carbonyl group, and the strong region near 1200 cm^{-1} is consistent with a C–O bond.

STEP 2 The absence of a broad alcohol or carboxylic-acid O–H absorption supports an ester.

DETAILED MARKING SCHEME

MARK	EVIDENCE REQUIRED	VALUE
Mark 1	Identifies the compound as an ester.	1 mark
Mark 2	Cites appropriate IR evidence: ester carbonyl/C–O absorption and absence of broad O–H.	1 mark

Award credit for an equivalent chemically valid method when the same evidence is demonstrated. A correct answer without required supporting working may not earn every mark.

PAPER 2 - QUESTION 1 - CONTINUED

(6 marks)

- b) Use the molecular-ion peak to determine the molecular formula.

[2 marks]

WORKED SOLUTION

STEP 1 The molecular-ion peak is at $m/z = 88$, so the molar mass is approximately 88 g mol^{-1} .

STEP 2 The empirical-formula mass of $\text{C}_2\text{H}_4\text{O}$ is 44 g mol^{-1} . The multiplier is $88/44 = 2$, so the molecular formula is $\text{C}_4\text{H}_8\text{O}_2$.

DETAILED MARKING SCHEME

MARK	EVIDENCE REQUIRED	VALUE
Mark 1	Uses $m/z = 88$ and empirical-formula mass 44 to obtain multiplier 2.	1 mark
Mark 2	Determines $\text{C}_4\text{H}_8\text{O}_2$.	1 mark

Award credit for an equivalent chemically valid method when the same evidence is demonstrated. A correct answer without required supporting working may not earn every mark.

- c) Deduce the structural formula and IUPAC name of the liquid. Explain how the synthesis information supports your answer.

[2 marks]

WORKED SOLUTION

STEP 1 An ester with formula $\text{C}_4\text{H}_8\text{O}_2$ made from methanol must contain the $-\text{OCH}_3$ group. The remaining acyl fragment is propanoyl, consistent with the strong $m/z = 57$ fragment.

STEP 2 The structure is $\text{CH}_3\text{CH}_2\text{COOCH}_3$ and the name is methyl propanoate.

DETAILED MARKING SCHEME

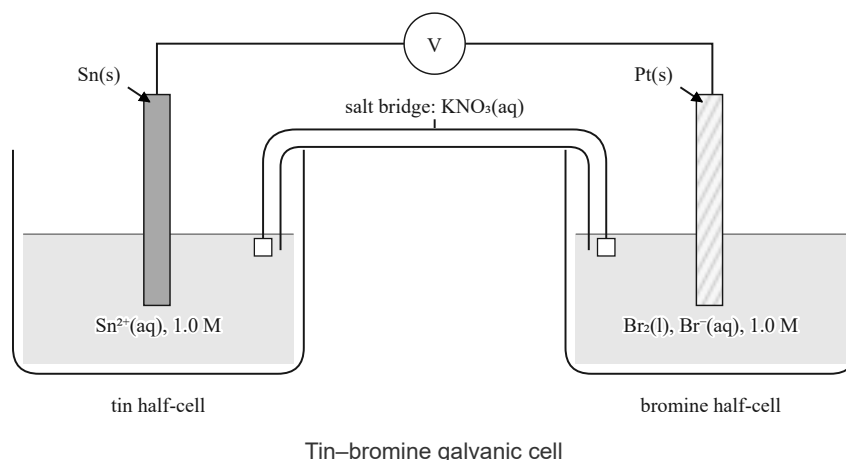
MARK	EVIDENCE REQUIRED	VALUE
Mark 1	Deduces the structure $\text{CH}_3\text{CH}_2\text{COOCH}_3$ from the combined evidence.	1 mark
Mark 2	Names the compound methyl propanoate.	1 mark

Award credit for an equivalent chemically valid method when the same evidence is demonstrated. A correct answer without required supporting working may not earn every mark.

PAPER 2 - QUESTION 2

(7 marks)

A galvanic cell uses a tin half-cell and a bromine/bromide half-cell under standard conditions. The inert electrode in the bromine half-cell is platinum. The standard reduction potentials are $E^\circ(\text{Br}_2/\text{Br}^-) = +1.08 \text{ V}$ and $E^\circ(\text{Sn}^{2+}/\text{Sn}) = -0.14 \text{ V}$.



- a) Write the oxidation and reduction half-equations. Include states. [2 marks]

WORKED SOLUTION

STEP 1 Oxidation: $\text{Sn(s)} \rightarrow \text{Sn}^{2+}(\text{aq}) + 2\text{e}^-$.

STEP 2 Reduction: $\text{Br}_2(\text{l}) + 2\text{e}^- \rightarrow 2\text{Br}^-(\text{aq})$.

DETAILED MARKING SCHEME

MARK	EVIDENCE REQUIRED	VALUE
Mark 1	Writes the balanced tin oxidation half-equation with states.	1 mark
Mark 2	Writes the balanced bromine reduction half-equation with states.	1 mark

Award credit for an equivalent chemically valid method when the same evidence is demonstrated. A correct answer without required supporting working may not earn every mark.

- b) State the direction of electron flow and the direction in which anions from the salt bridge initially move. [2 marks]

WORKED SOLUTION

STEP 1 Electrons flow through the wire from the tin electrode to the platinum electrode.

STEP 2 Oxidation produces Sn^{2+} in the tin half-cell, so salt-bridge anions move towards the tin half-cell to maintain electrical neutrality.

DETAILED MARKING SCHEME

MARK	EVIDENCE REQUIRED	VALUE
Mark 1	States electron flow from tin to platinum/bromine half-cell.	1 mark
Mark 2	States that salt-bridge anions move to the tin half-cell and links this to charge balance.	1 mark

Award credit for an equivalent chemically valid method when the same evidence is demonstrated. A correct answer without required supporting working may not earn every mark.

PAPER 2 - QUESTION 2 - CONTINUED

(7 marks)

- c) Calculate the standard cell potential.

[1 mark]

WORKED SOLUTION

STEP 1 $E_{\text{cell}}^{\circ} = 1.08 - (-0.14) = +1.22 \text{ V}.$

DETAILED MARKING SCHEME

MARK	EVIDENCE REQUIRED	VALUE
Mark 1	Obtains $E_{\text{cell}}^{\circ} = +1.22 \text{ V}.$	1 mark

Award credit for an equivalent chemically valid method when the same evidence is demonstrated. A correct answer without required supporting working may not earn every mark.

- d) Describe two observable changes as the cell operates.

[2 marks]

WORKED SOLUTION

STEP 1 The tin electrode loses mass as tin atoms oxidise to Sn^{2+} .

STEP 2 The amount (or volume) of red-brown liquid bromine decreases as $\text{Br}_2(\text{l})$ is reduced to colourless bromide ions.

DETAILED MARKING SCHEME

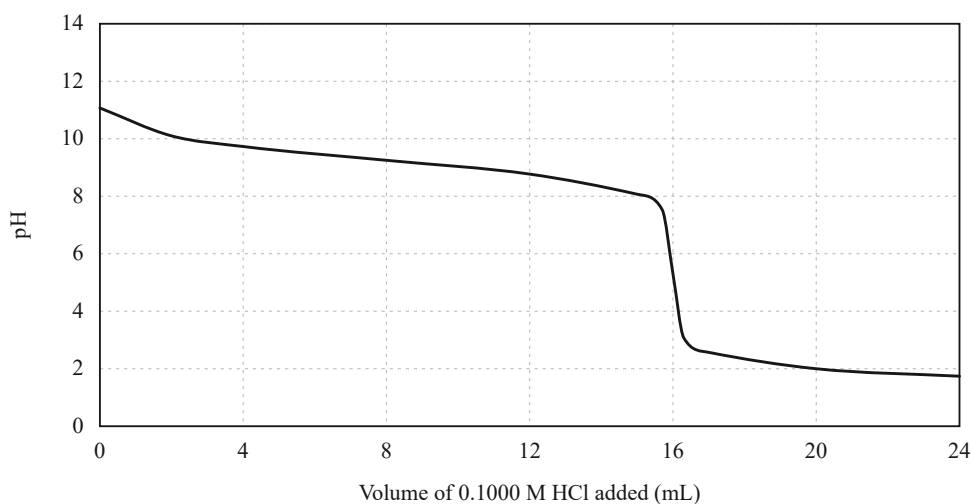
MARK	EVIDENCE REQUIRED	VALUE
Mark 1	Describes loss of mass at the tin electrode.	1 mark
Mark 2	Describes a decrease in the amount or volume of red-brown liquid bromine in the bromine half-cell.	1 mark

Award credit for an equivalent chemically valid method when the same evidence is demonstrated. A correct answer without required supporting working may not earn every mark.

PAPER 2 - QUESTION 3

(6 marks)

At 25 °C, 20.00 mL of an ammonia solution is titrated with 0.1000 M hydrochloric acid. The titration curve has an equivalence volume of 16.00 mL and a pH of 9.25 at half-equivalence.



Ammonia–hydrochloric acid titration curve

Indicator	Transition range
methyl red	4.4-6.2
bromothymol blue	6.0-7.6
phenolphthalein	8.3-10.0

Indicator transition ranges

- a) Calculate the initial concentration of ammonia.

[2 marks]

WORKED SOLUTION

STEP 1 At equivalence, $n(\text{HCl}) = (0.1000)(0.01600) = 1.600 \times 10^{-3} \text{ mol}$, equal to the initial moles of ammonia.

STEP 2 $[\text{NH}_3] = (1.600 \times 10^{-3})/0.02000 = 0.08000 \text{ M}$.

DETAILED MARKING SCHEME

MARK	EVIDENCE REQUIRED	VALUE
Mark 1	Calculates $1.600 \times 10^{-3} \text{ mol}$ at equivalence.	1 mark
Mark 2	Obtains $[\text{NH}_3] = 0.08000 \text{ M}$.	1 mark

Award credit for an equivalent chemically valid method when the same evidence is demonstrated. A correct answer without required supporting working may not earn every mark.

PAPER 2 - QUESTION 3 - CONTINUED

(6 marks)

- b) Use the half-equivalence point to determine K_b for ammonia.

[2 marks]

WORKED SOLUTION

STEP 1 At half-equivalence, $\text{pOH} = \text{p}K_b$. Thus $\text{p}K_b = 14.00 - 9.25 = 4.75$.

STEP 2 $K_b = 10^{-4.75} = 1.8 \times 10^{-5}$.

DETAILED MARKING SCHEME

MARK	EVIDENCE REQUIRED	VALUE
Mark 1	Determines $\text{p}K_b = 4.75$.	1 mark
Mark 2	Obtains $K_b = 1.8 \times 10^{-5}$.	1 mark

Award credit for an equivalent chemically valid method when the same evidence is demonstrated. A correct answer without required supporting working may not earn every mark.

- c) Select the most suitable indicator from the table.

[1 mark]

WORKED SOLUTION

STEP 1 Methyl red is suitable because its transition range, pH 4.4-6.2, lies within the steep region around the acidic equivalence point.

DETAILED MARKING SCHEME

MARK	EVIDENCE REQUIRED	VALUE
Mark 1	Selects methyl red.	1 mark

Award credit for an equivalent chemically valid method when the same evidence is demonstrated. A correct answer without required supporting working may not earn every mark.

- d) The burette was rinsed with water immediately before it was filled with hydrochloric acid. State whether using the labelled acid concentration makes the calculated ammonia concentration too high or too low.

[1 mark]

WORKED SOLUTION

STEP 1 Residual water dilutes the acid, so a larger titre is required. Using that larger volume with the undiluted labelled concentration overestimates the moles of acid and therefore overestimates the ammonia concentration.

DETAILED MARKING SCHEME

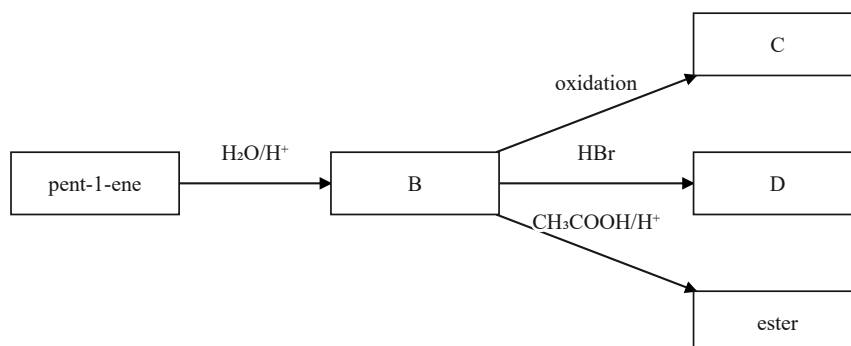
MARK	EVIDENCE REQUIRED	VALUE
Mark 1	Predicts an overestimated ammonia concentration with a correct dilution-to-titre explanation.	1 mark

Award credit for an equivalent chemically valid method when the same evidence is demonstrated. A correct answer without required supporting working may not earn every mark.

PAPER 2 - QUESTION 4

(7 marks)

Use the displayed reaction pathway, which starts with pent-1-ene. Compound B is the major hydration product.



Reaction pathway

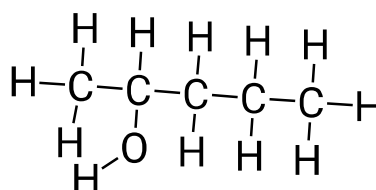
- a) Draw the structural formula of B and give its IUPAC name.

[2 marks]

WORKED SOLUTION

STEP 1 Markovnikov hydration places the hydroxyl group on carbon 2.

STEP 2 B is $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_2\text{CH}_3$, pentan-2-ol.



Compound B — pentan-2-ol

DETAILED MARKING SCHEME

MARK	EVIDENCE REQUIRED	VALUE
Mark 1	Draws $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_2\text{CH}_3$ with correct connectivity.	1 mark
Mark 2	Names B as pentan-2-ol.	1 mark

Award credit for an equivalent chemically valid method when the same evidence is demonstrated. A correct answer without required supporting working may not earn every mark.

PAPER 2 - QUESTION 4 - CONTINUED

(7 marks)

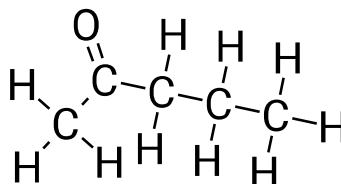
- b) State a reagent for the conversion of B to C and draw the structural formula of C.

[2 marks]

WORKED SOLUTION

STEP 1 Use acidified potassium dichromate(VI) or acidified potassium manganate(VII).

STEP 2 The secondary alcohol oxidises to $\text{CH}_3\text{COCH}_2\text{CH}_2\text{CH}_3$, pentan-2-one.



Compound C — pentan-2-one

DETAILED MARKING SCHEME

MARK	EVIDENCE REQUIRED	VALUE
Mark 1	States a valid acidified oxidising reagent.	1 mark
Mark 2	Draws the ketone $\text{CH}_3\text{COCH}_2\text{CH}_2\text{CH}_3$.	1 mark

Award credit for an equivalent chemically valid method when the same evidence is demonstrated. A correct answer without required supporting working may not earn every mark.

- c) Classify the conversion of pent-1-ene to B and the conversion of B to D.

[2 marks]

WORKED SOLUTION

STEP 1 Alkene hydration is an addition reaction.

STEP 2 Replacement of the alcohol hydroxyl group by bromine using HBr is a substitution reaction.

DETAILED MARKING SCHEME

MARK	EVIDENCE REQUIRED	VALUE
Mark 1	Classifies alkene hydration as addition.	1 mark
Mark 2	Classifies alcohol-to-haloalkane conversion as substitution.	1 mark

Award credit for an equivalent chemically valid method when the same evidence is demonstrated. A correct answer without required supporting working may not earn every mark.

PAPER 2 - QUESTION 4 - CONTINUED

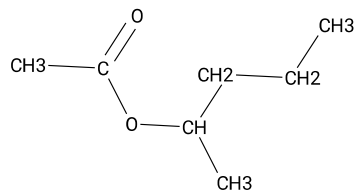
(7 marks)

- d) Draw the ester formed when B reacts with ethanoic acid.

[1 mark]

WORKED SOLUTION

STEP 1 The ester is $\text{CH}_3\text{COOCH}(\text{CH}_3)\text{CH}_2\text{CH}_2\text{CH}_3$.



Ester product — pentan-2-yl ethanoate

DETAILED MARKING SCHEME

MARK	EVIDENCE REQUIRED	VALUE
Mark 1	Draws $\text{CH}_3\text{COOCH}(\text{CH}_3)\text{CH}_2\text{CH}_2\text{CH}_3$ with the ester linkage and correct alcohol-derived group.	1 mark

Award credit for an equivalent chemically valid method when the same evidence is demonstrated. A correct answer without required supporting working may not earn every mark.

PAPER 2 - QUESTION 5

(6 marks)

0.800 mol of $\text{PCl}_5(\text{g})$ is placed in a 2.00 L vessel. At equilibrium, $[\text{Cl}_2] = 0.150 \text{ M}$ for $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$.

- a) Determine the equilibrium concentrations of PCl_5 and PCl_3 . [2 marks]

WORKED SOLUTION

STEP 1 0.150 M of Cl_2 corresponds to 0.300 mol formed, so the same amount of PCl_3 forms and 0.300 mol of PCl_5 is consumed.

STEP 2 $[\text{PCl}_3] = 0.150 \text{ M}$ and $[\text{PCl}_5] = (0.800 - 0.300)/2.00 = 0.250 \text{ M}$.

DETAILED MARKING SCHEME

MARK	EVIDENCE REQUIRED	VALUE
Mark 1	Determines $[\text{PCl}_3] = 0.150 \text{ M}$.	1 mark
Mark 2	Determines $[\text{PCl}_5] = 0.250 \text{ M}$.	1 mark

Award credit for an equivalent chemically valid method when the same evidence is demonstrated. A correct answer without required supporting working may not earn every mark.

- b) Calculate K_c . [2 marks]

WORKED SOLUTION

STEP 1 $K_c = \frac{[\text{PCl}_3][\text{Cl}_2]}{[\text{PCl}_5]}$.

STEP 2 $K_c = (0.150)(0.150)/(0.250) = 0.0900$.

DETAILED MARKING SCHEME

MARK	EVIDENCE REQUIRED	VALUE
Mark 1	Writes/substitutes into the correct equilibrium expression.	1 mark
Mark 2	Obtains $K_c = 0.0900$.	1 mark

Award credit for an equivalent chemically valid method when the same evidence is demonstrated. A correct answer without required supporting working may not earn every mark.

- c) At the same temperature, a second mixture has $[\text{PCl}_5] = 0.200 \text{ M}$, $[\text{PCl}_3] = 0.100 \text{ M}$ and $[\text{Cl}_2] = 0.300 \text{ M}$. Infer the direction of the net reaction. [2 marks]

WORKED SOLUTION

STEP 1 $Q = (0.100)(0.300)/(0.200) = 0.150$.

STEP 2 Since $Q > K_c$, there is too much product relative to equilibrium and the net reaction proceeds left.

DETAILED MARKING SCHEME

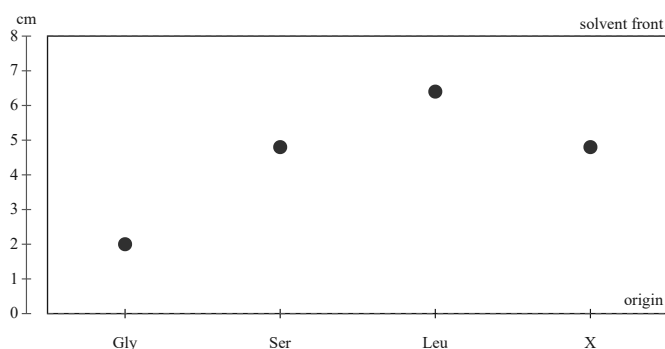
MARK	EVIDENCE REQUIRED	VALUE
Mark 1	Calculates $Q = 0.150$.	1 mark
Mark 2	Infers a net reverse/left reaction because $Q > K_c$.	1 mark

Award credit for an equivalent chemically valid method when the same evidence is demonstrated. A correct answer without required supporting working may not earn every mark.

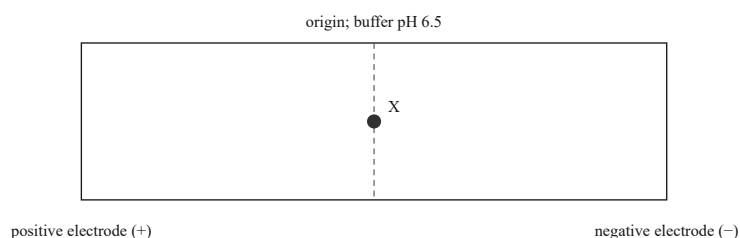
PAPER 2 - QUESTION 6

(6 marks)

An unknown amino acid X is analysed using the paper chromatogram and electrophoresis arrangement shown. The chromatography uses polar cellulose paper with adsorbed water as the stationary phase and a less-polar organic solvent as the mobile phase. Serine has an isoelectric point of 5.68.



Paper chromatogram



Electrophoresis at pH 6.50

- a) Calculate the R_f value of X and identify it.

[2 marks]

WORKED SOLUTION

STEP 1 $R_f = 4.8/8.0 = 0.60$.

STEP 2 The value matches the serine standard, so X is serine.

DETAILED MARKING SCHEME

MARK	EVIDENCE REQUIRED	VALUE
Mark 1	Calculates $R_f = 0.60$.	1 mark
Mark 2	Identifies X as serine.	1 mark

Award credit for an equivalent chemically valid method when the same evidence is demonstrated. A correct answer without required supporting working may not earn every mark.

PAPER 2 - QUESTION 6 - CONTINUED

(6 marks)

- b) Leucine moves further than glycine in the stated paper-chromatography system. Explain this difference in terms of the stationary and mobile phases. [2 marks]

WORKED SOLUTION

STEP 1 The polar paper/water stationary phase interacts more strongly with the more polar glycine, retaining it closer to the origin.

STEP 2 Leucine's less-polar side chain gives it greater relative affinity/solubility in the mobile phase, so it travels further.

DETAILED MARKING SCHEME

MARK	EVIDENCE REQUIRED	VALUE
Mark 1	Links glycine's polarity to stronger stationary-phase interaction/retention.	1 mark
Mark 2	Links leucine's lower polarity to greater mobile-phase movement.	1 mark

Award credit for an equivalent chemically valid method when the same evidence is demonstrated. A correct answer without required supporting working may not earn every mark.

- c) Predict the net charge of X and its direction of movement in the pH 6.50 buffer. [2 marks]

WORKED SOLUTION

STEP 1 The buffer pH is above serine's isoelectric point, so serine has a net negative charge.

STEP 2 It moves towards the positive electrode.

DETAILED MARKING SCHEME

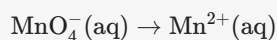
MARK	EVIDENCE REQUIRED	VALUE
Mark 1	Determines that serine has a net negative charge at pH 6.50.	1 mark
Mark 2	Predicts movement towards the positive electrode/anode.	1 mark

Award credit for an equivalent chemically valid method when the same evidence is demonstrated. A correct answer without required supporting working may not earn every mark.

PAPER 2 - QUESTION 7

(6 marks)

Acidified permanganate ions oxidise iron(II) ions to iron(III) ions. A 25.00 mL iron(II) sample requires 18.60 mL of 0.02000 M permanganate solution for complete reaction.



Reduction skeleton

- a) Write the oxidation half-equation and the balanced reduction half-equation in acidic conditions. [3 marks]

WORKED SOLUTION

STEP 1 Oxidation: $\text{Fe}^{2+} (\text{aq}) \rightarrow \text{Fe}^{3+} (\text{aq}) + \text{e}^-$.

STEP 2 Reduction: $\text{MnO}_4^- (\text{aq}) + 8\text{H}^+ (\text{aq}) + 5\text{e}^- \rightarrow \text{Mn}^{2+} (\text{aq}) + 4\text{H}_2\text{O} (\text{l})$.

DETAILED MARKING SCHEME

MARK	EVIDENCE REQUIRED	VALUE
Mark 1	Writes the iron(II) oxidation half-equation.	1 mark
Mark 2	Balances oxygen/hydrogen in the permanganate reduction with 8H^+ and $4\text{H}_2\text{O}$.	1 mark
Mark 3	Balances charge with 5e^- and includes correct states.	1 mark

Award credit for an equivalent chemically valid method when the same evidence is demonstrated. A correct answer without required supporting working may not earn every mark.

- b) Calculate the concentration of iron(II) ions in the sample. [3 marks]

WORKED SOLUTION

STEP 1 $n(\text{MnO}_4^-) = (0.02000)(0.01860) = 3.720 \times 10^{-4} \text{ mol}$.

STEP 2 The overall ratio is $1 \text{ MnO}_4^- : 5 \text{ Fe}^{2+}$, so $n(\text{Fe}^{2+}) = 1.860 \times 10^{-3} \text{ mol}$.

STEP 3 $[\text{Fe}^{2+}] = (1.860 \times 10^{-3}) / 0.02500 = 0.07440 \text{ M}$.

DETAILED MARKING SCHEME

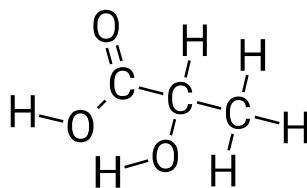
MARK	EVIDENCE REQUIRED	VALUE
Mark 1	Calculates $3.720 \times 10^{-4} \text{ mol}$ of permanganate.	1 mark
Mark 2	Uses the 1 : 5 ratio to obtain $1.860 \times 10^{-3} \text{ mol}$ of iron(II).	1 mark
Mark 3	Obtains $[\text{Fe}^{2+}] = 0.07440 \text{ M}$.	1 mark

Award credit for an equivalent chemically valid method when the same evidence is demonstrated. A correct answer without required supporting working may not earn every mark.

PAPER 2 - QUESTION 8

(6 marks)

The displayed lactic acid monomer can form polylactic acid (PLA) by condensation polymerisation.



Lactic acid monomer

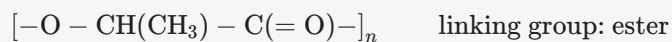
- a) Draw the repeating unit of PLA and identify the linking functional group.

[2 marks]

WORKED SOLUTION

STEP 1 The repeating unit is $[-OCH(CH_3)CO-]_n$.

STEP 2 The units are joined by ester linkages.



PLA repeating unit

DETAILED MARKING SCHEME

MARK	EVIDENCE REQUIRED	VALUE
Mark 1	Draws $[-OCH(CH_3)CO-]_n$ with brackets and continuation bonds.	1 mark
Mark 2	Identifies the ester linkage.	1 mark

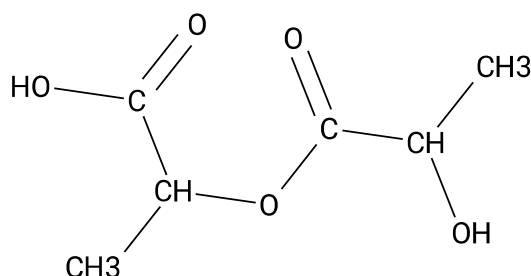
Award credit for an equivalent chemically valid method when the same evidence is demonstrated. A correct answer without required supporting working may not earn every mark.

PAPER 2 - QUESTION 8 - CONTINUED

(6 marks)

- b) Write an equation for two lactic acid molecules forming a dimer. Show the small molecule eliminated. [2 marks]

WORKED SOLUTION



Lactic-acid condensation dimer

DETAILED MARKING SCHEME

MARK	EVIDENCE REQUIRED	VALUE
Mark 1	Draws/writes the lactic-acid dimer with one ester bond.	1 mark
Mark 2	Shows one water molecule eliminated and balances the equation.	1 mark

Award credit for an equivalent chemically valid method when the same evidence is demonstrated. A correct answer without required supporting working may not earn every mark.

- c) Explain why PLA is more susceptible to hydrolysis and biodegradation than polyethene. [2 marks]

WORKED SOLUTION

STEP 1 PLA contains polar ester linkages in its backbone that can undergo hydrolysis, allowing chain cleavage.

STEP 2 Polyethene has a nonpolar carbon–carbon backbone without hydrolysable functional groups, so it is less susceptible.

DETAILED MARKING SCHEME

MARK	EVIDENCE REQUIRED	VALUE
Mark 1	Links PLA ester groups to hydrolysis/chain cleavage.	1 mark
Mark 2	Contrasts the nonpolar, non-hydrolysable polyethene backbone.	1 mark

Award credit for an equivalent chemically valid method when the same evidence is demonstrated. A correct answer without required supporting working may not earn every mark.

PAPER 2 - QUESTION 9

(5 marks)

Four Haber-process trials begin with the same reactant ratio. The percentage ammonia at equilibrium and the time taken to approach equilibrium are recorded.

Trial	Temperature (°C)	Pressure (atm)	Catalyst	NH ₃ at equilibrium	Time to approach equilibrium (min)
A	400	100	yes	25%	20
B	450	100	yes	18%	12
C	450	200	yes	28%	8
D	450	200	no	28%	65

Haber-process trial data

- a) Use trials A and B to infer whether the forward reaction is exothermic or endothermic. Explain your reasoning. [2 marks]

WORKED SOLUTION

STEP 1 Raising temperature from 400 to 450 °C lowers the equilibrium ammonia percentage from 25% to 18%.

STEP 2 Heating therefore favours the reverse direction, so the forward production of ammonia is exothermic.

DETAILED MARKING SCHEME

MARK	EVIDENCE REQUIRED	VALUE
Mark 1	Identifies the decrease in equilibrium ammonia percentage as temperature rises.	1 mark
Mark 2	Infers that the forward reaction is exothermic.	1 mark

Award credit for an equivalent chemically valid method when the same evidence is demonstrated. A correct answer without required supporting working may not earn every mark.

- b) Use trials C and D to explain the effect of the catalyst on rate and equilibrium yield. [2 marks]

WORKED SOLUTION

STEP 1 The catalyst reduces the time to approach equilibrium from 65 to 8 minutes, so it increases the reaction rates.

STEP 2 Both trials reach 28% ammonia, so the catalyst does not change the equilibrium composition or K_c at the same temperature.

DETAILED MARKING SCHEME

MARK	EVIDENCE REQUIRED	VALUE
Mark 1	Uses the shorter time to infer a faster approach to equilibrium.	1 mark
Mark 2	Uses the equal equilibrium percentages to conclude that the catalyst does not change yield/ K_c .	1 mark

Award credit for an equivalent chemically valid method when the same evidence is demonstrated. A correct answer without required supporting working may not earn every mark.

PAPER 2 - QUESTION 9 - CONTINUED

(5 marks)

- c) Identify the condition change responsible for the higher equilibrium ammonia percentage in trial C than trial B. [1 mark]

WORKED SOLUTION

STEP 1 Pressure is increased from 100 to 200 atm, favouring the product side with fewer gas particles.

DETAILED MARKING SCHEME

MARK	EVIDENCE REQUIRED	VALUE
Mark 1	Identifies increased pressure and connects it to higher ammonia yield.	1 mark

Award credit for an equivalent chemically valid method when the same evidence is demonstrated. A correct answer without required supporting working may not earn every mark.